



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

MA2001-1 – STATISTICS AND PROBABILITY THEORY
 (For AD, AI, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2) Statistical table permitted.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- If $P(A) = 1/4$, $P(B) = 1/3$, $P(A \cup B) = 1/2$ then $P(A|B)$ is _____
 A) $1/12$ B) $1/3$
 C) $1/4$ D) $1/6$
- If X is a discrete random variable with $(x, p(x))$ values as $(0, k), (1, 2k), (2, 3k), (3, 4k)$, then value of k is _____
 A) 0.4 B) 10
 C) 0.1 D) 1
- If $E[X] = 2$, $E[X^2] = 12$ then standard deviation is _____
 A) 8 B) 2.8284
 C) 0.2828 D) 3.1622
- The pdf of random variable is $2x$, $0 < x < 1$ then $P(0 < X < 0.5)$ is _____
 A) 0.125 B) 0.25
 C) 0.5 D) 1
- In binomial distribution if $n=10$ and $p=0.1$ then standard deviation is _____
 A) 9 B) 0.9
 C) 0.9486 D) 1
- The pdf of standard normal distribution is symmetric about _____
 A) x-axis B) y-axis
 C) About the line $x = \sigma$ D) Both (A) and (B)
- In Poisson distribution if $n=10000$ and $p=0.001$ then standard deviation is _____
 A) 10 B) 3.4641
 C) 2.8284 D) 3.1622
- In exponential distribution if $\alpha = 1/2$ then variance is _____
 A) $\sqrt{4}$ B) $1/4$
 C) 4 D) $1/\sqrt{4}$
- If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2 , then the limiting form of the distribution of $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ as $n \rightarrow \infty$ is _____
 A) Standard normal distribution B) Chi-square distribution
 C) t-distribution D) Exponential distribution
- Let $\hat{\theta}$ be an estimate for the unknown parameter θ associated with the distribution of random variable X . If $E(\hat{\theta}) = \theta$, for all, then $\hat{\theta}$ is _____
 A) Biased estimator B) Mean
 C) Variance D) unbiased estimator

11. When sample size $n < 30$, we use----- distribution
 A) Normal distribution B) t-distribution
 C) Chi-square distribution D) Exponential distribution
12. If $X \sim N(\mu, \sigma^2)$ where μ and σ^2 are unknown. Then which of the following is a statistic?
 A) $X^2 + 5X$ B) $X + \mu$
 C) $X^2 - \sigma^2$ D) $\frac{X^2}{\sigma^2}$
13. If the regression lines are $x = 0.7y + 5.2$ and $y = 0.3x + 2.8$
 then $(\bar{x}, \bar{y})$ is _____
 A) (0.7, 0.3) B) (5.2, 2.8)
 C) (1, 1) D) (9, 5.52)
14. Which statement is not correct?
 A) Coefficient of correlation is the arithmetic mean between the regression coefficients
 B) Coefficient of correlation is the geometric mean between the regression coefficients
 C) Coefficient of correlation ρ lies in $[-1, 1]$.
 D) Coefficient of correlation ρ is independent of origin
15. The standard deviation of x and y are 6 and 8 respectively. The correlation coefficient between them is 0.6. Then the value of regression coefficient b_{yx} is _____
 A) 0.6 B) 0.75
 C) 0.45 D) 0.8
16. One of the normal equations for fitting of a straight line $y = a + bx$ is $\sum y =$ _____
 A) $na + b \sum x$ B) $na + b \sum x^2$
 C) $n^2a + b \sum x$ D) $a + b \sum x$
17. Which of the following is not a Probability vector.
 A) $(\frac{1}{3}, \frac{2}{3}, 0)$ B) $(\frac{1}{3}, \frac{-1}{3}, \frac{2}{3})$
 C) $(\frac{1}{5}, \frac{2}{5}, \frac{2}{5})$ D) $(0, \frac{3}{5}, \frac{2}{5})$
18. Which of the following best describes queuing theory?
 A) The study of arrival rates
 B) The evaluation of service time costs
 C) The study of service times
 D) The study of waiting lines
19. Customer behaviour in which he moves from one queue to another in multiple channel situation is _____
 A) Balking B) Jockeying
 C) Alternating D) Reneging
20. Which of the following is a stochastic matrix
 A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ B) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$
 C) $\begin{bmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{2} \end{bmatrix}$ D) $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

PART – B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a)	x	-2	-1	0	1	2	3
	P(x)	0.1	k	0.2	2k	0.3	k

The following table gives the probability distribution of a random variable:

Find i) the value of k (ii) $P(X > 0)$ (iii) the Mean and Standard deviation.

Marks BT* CO* PO*

6 L*2 1 1,2

- b) Two random variables X and Y have joint pdf

$$f(x, y) = \begin{cases} \frac{1}{3}(4 - x - y), & 1 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Compute $E(X)$, $E(Y)$, $E(XY)$

5 L2 1 1,2

- c) The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that i) exactly two will be defective ii) at least two will be defective iii) none will be defective.

5 L2 2 1,2

2. a) A random variable X has the probability density function $f(x) =$

$$\begin{cases} kx^2 & -3 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find (i) The value of k (ii) $P(X < 1.5)$ (iii) cdf of X

6 L2 1 1,2

- b) If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals (i) exactly 3 (ii) more than two get a bad reaction.

5 L2 2 1,2

- c) Obtain the mean and standard deviation of uniform distribution

5 L2 2 1,2

- 3 a) The joint probability distribution of (X, Y) is given by the following table

Y \ X	X		
	-3	2	4
1	0.1	0.2	0.2
3	0.3	0.1	0.1

Find (i) the marginal distribution of X and Y

(ii) the mean of X and Y

(iii) Are X and Y independent?

6 L2 1 1,2

- b) Suppose that X has poisson distribution

(i) Determine the mgf of X.

(ii) Using mgf, evaluate $E(X)$ and $V(X)$.

5 L2 2 1,2

- c) The life length of an instrument produced by a machine has a normal distribution with a mean of 12 months and a standard deviation of 2 months. Find the probability that an instrument produced by this machine will last

(i) less than 10 months

(ii) between 11 and 14 months.

5 L2 2 1,2

Unit - II

4. a) State Central limit theorem. Let $\bar{x}$ denote the mean of a random sample of size n from a distribution which has mean μ and $\sigma^2 = 10$. Find n if the probability that the interval $(\bar{x} - 1/2, \bar{x} + 1/2)$ contains μ is 0.954.

6 L2 3 1,2

- b) Compute the rank correlation coefficient for the following data:

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

5 L2 4 1,2

- c) By the method of least squares, find the straight line $y = a + bx$ that best fits the following data:

x	1	2	3	4	5
y	14	27	40	55	68

5 L2 4 1,2

5. a) If $\bar{X}$ is the mean of a random sample of size n from a population with mean μ , variance σ^2 and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, then prove that $E(\bar{X}) = \mu$ and $E(S^2) = \sigma^2$

6 L2 3 1,2

- b) Find the correlation coefficient between x and y for the given values.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

5 L2 4 1,2

- c) Let S^2 be the variance of a random sample of size 6 from a normal distribution $N(\mu, 12)$, find $P(2.3 < S^2 < 22.2)$

5 L2 3 1,2

6. a) Let a random sample of size 17 from a normal distribution $N(\mu, \sigma^2)$ yield $\bar{X} = 4.7, S^2 = 5.76$. Determine a 90% confidence interval for μ and σ^2

6 L2 3 1,2

- b) The equation of regression lines of two variables x and y are $x = 19.13 - 0.87y$ and $y = 11.64 - 0.50x$. Find the correlation coefficient and means of x and y .

5 L2 4 1,2

- c) Fit a quadratic curve $y = ax^2 + bx + c$ to the data using the least square method

x	5	10	15	20	25
y	21	24	33	42	38

5 L2 4 1,2

Unit – III

7. a) (i) If $A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$ is a stochastic matrix and $V = [v_1 \ v_2]$ is a probability vector. Show that VA is also a probability vector.

(ii) Prove that with reference to the two second order stochastic matrices that their product is also a stochastic matrix.

6 L2 5 1,2

- b) Consider a Markov Chain with three states $S = \{1, 2, 3\}$ and transition

$$\text{matrix } P = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

(i) Draw the state transition diagram for this chain.

(ii) If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

5 L2 5 1,2

- c) Define regular stochastic matrix. Verify that the matrix $A =$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/4 & 1/4 \\ 0 & 1 & 0 \end{bmatrix} \text{ is a regular stochastic.}$$

5 L2 5 1,2

8. a) A software engineer goes to his workplace every day either by bike or by car. He never goes by bike on two consecutive days; but if he goes by car on a day then he is equally likely to go by car or bike on the next day. Find the transition matrix for the chain of the mode of transport he uses. If car is used on the first day of the week, find the probability that after four days (i) bike is used (ii) car is used.

6 L2 5 1,2

- b) Find the unique fixed probability vector for the regular stochastic

$$\text{matrix } A = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$$

5 L2 5 1,2